

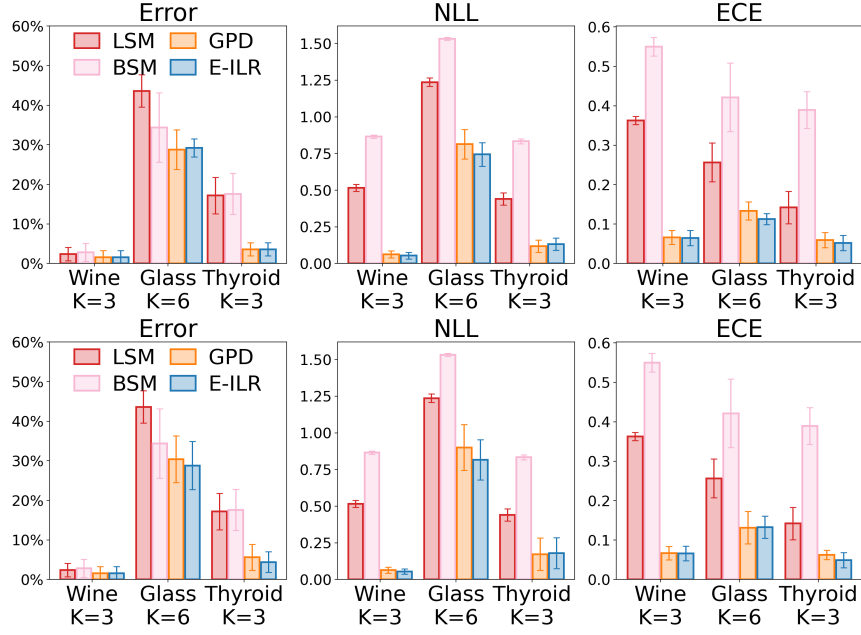


Figure A: Error, NLL and ECE for UCI datasets in the exact setting. **Top row:** Results shown in the paper. **Bottom row:** For each of the five splits, hyperparameters are selected using that split’s validation set.

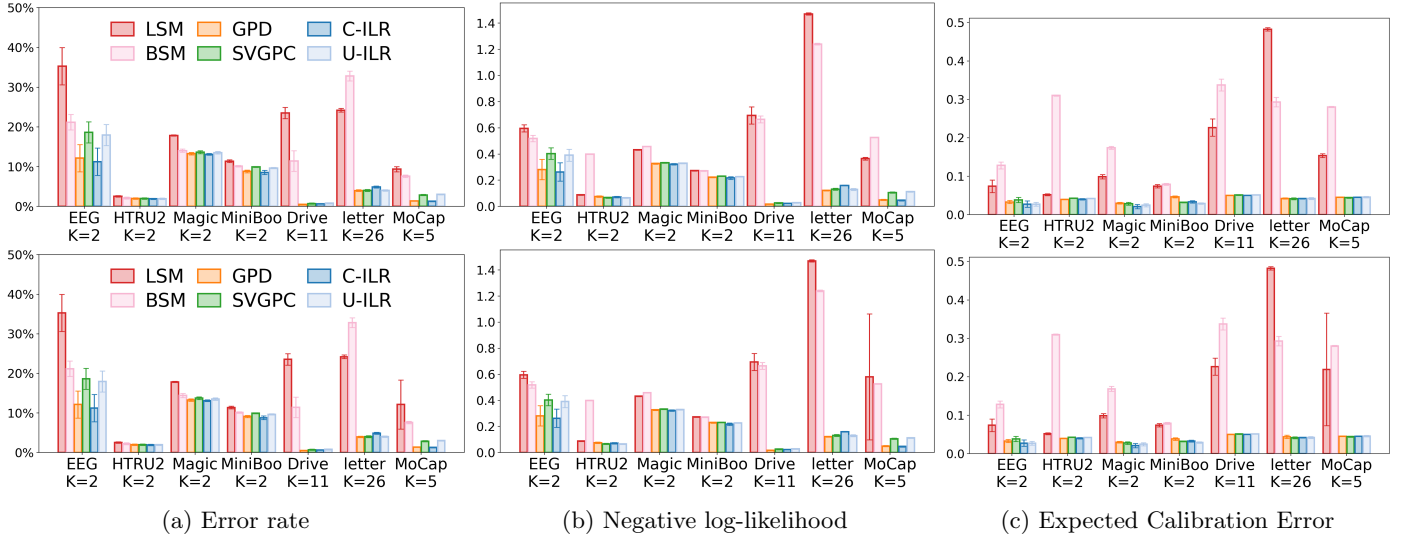


Figure B: Error, NLL and ECE for UCI datasets in the sparse setting. **Top row:** Results shown in the paper. **Bottom row:** For each of the five splits, hyperparameters are selected using that split’s validation set.

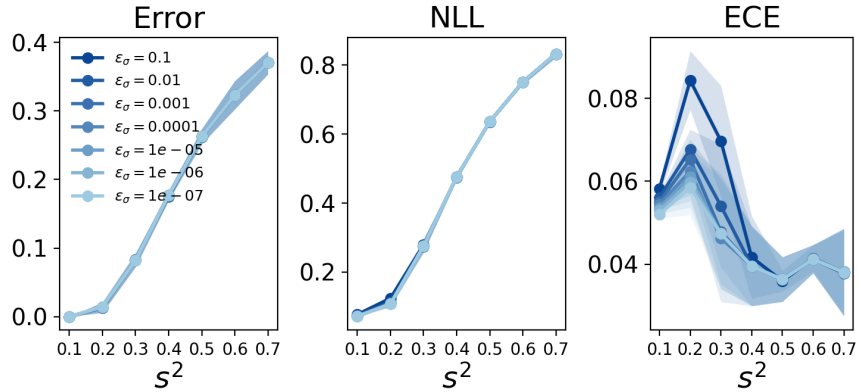


Figure C: Effect of σ . We replicate the setup of “Effect of the parameter λ ” (See Fig. 5). We fix $\lambda = 0.9$ and vary $\epsilon_\sigma \in \{0.1, \dots, 10^{-7}\}$. The parameter σ set based the lower bound dependent on ϵ_σ . For $\epsilon \geq 0.01$ the method is stable to different values of σ .

Table A: Full details on the split-wise hyperparameter selection for each model and experiment. For each of the five splits, hyperparameters are selected using that split’s validation set, and Accuracy, NLL, and ECE are then computed on the corresponding test set. Reported values are means and standard deviations over the five test splits, with the best mean value boldfaced. Norm. refers to the input normalization method and LR to the learning rate. Time is the average time in seconds of the parameter fit. The execution times of GPD and E-ILR are of the same order of magnitude, as are those of sparse GPD and C-ILR, and of SVGPC and U-ILR

Exp	Model	Norm	Target HP	LR	ACC $\uparrow$	NLL $\downarrow$	ECE $\downarrow$	Time(s)
Wine	LSM	$[-1, 1]$	–	0.001	0.98 ± 0.017	0.52 ± 0.023	0.36 ± 0.010	–
	BSM	$[-1, 1]$	–	0.001	0.97 ± 0.023	0.87 ± 0.011	0.55 ± 0.024	–
	GPD	$[-1, 1], Z$	$\alpha_E: 0.0001$	0.01	0.98 ± 0.017	0.06 ± 0.021	0.07 ± 0.017	5.9
	E-ILR (ours)	$[-1, 1], Z$	$\lambda: 0.999999$	0.01	0.98 ± 0.017	0.06 ± 0.018	0.07 ± 0.018	6.2
Glass	LSM	$[-1, 1]$	–	0.001	0.56 ± 0.041	1.24 ± 0.028	0.26 ± 0.049	–
	BSM	$[-1, 1]$	–	0.001	0.66 ± 0.088	1.53 ± 0.010	0.42 ± 0.087	–
	GPD	Z	$\alpha_E: 0.01, 0.0001, 0.001$	0.01, 0.001	0.70 ± 0.059	0.90 ± 0.157	0.13 ± 0.041	7.4
	E-ILR (ours)	Z, $[-1, 1]$	$\lambda: 0.99, 0.999999, 0.999, 0.9999, 0.99999$	0.001, 0.01	0.71 ± 0.061	0.82 ± 0.137	0.13 ± 0.028	7.7
Thyroid	LSM	$[-1, 1]$	–	0.001	0.83 ± 0.046	0.44 ± 0.042	0.14 ± 0.041	–
	BSM	$[-1, 1]$	–	0.001	0.82 ± 0.052	0.83 ± 0.017	0.39 ± 0.047	–
	GPD	$[-1, 1]$	$\alpha_E: 0.01, 0.001$	0.001, 0.01	0.94 ± 0.033	0.17 ± 0.111	0.06 ± 0.012	6.4
	E-ILR (ours)	$[-1, 1], Z$	$\lambda: 0.999, 0.99999, 0.9999, 0.999999, 0.9999999$	0.001, 0.01	0.96 ± 0.026	0.18 ± 0.106	0.05 ± 0.020	5.9
EEG	LSM	Z	–	0.01	0.65 ± 0.047	0.60 ± 0.027	0.07 ± 0.016	–
	BSM	Z	–	0.001	0.79 ± 0.019	0.52 ± 0.025	0.13 ± 0.009	–
	GPD	Z	$\alpha_E: 0.01$	0.01	0.88 ± 0.034	0.28 ± 0.077	0.03 ± 0.004	108
	SVGPC	Z	–	0.01	0.81 ± 0.027	0.40 ± 0.044	0.04 ± 0.007	29.4
	U-ILR (ours)	Z	–	0.01	0.82 ± 0.026	0.39 ± 0.046	0.03 ± 0.005	29.8
	C-ILR (ours)	Z	$\lambda: 0.99$	0.001	0.89 ± 0.034	0.26 ± 0.070	0.03 ± 0.008	56.5
HTRU2	LSM	$[-1, 1]$	–	0.01	0.97 ± 0.002	0.09 ± 0.004	0.05 ± 0.002	–
	BSM	$[-1, 1], Z$	–	0.001	0.98 ± 0.003	0.40 ± 0.003	0.31 ± 0.002	–
	GPD	$[-1, 1]$	$\alpha_E: 0.01$	0.01	0.98 ± 0.002	0.08 ± 0.006	0.04 ± 0.001	92.4
	SVGPC	$[-1, 1], Z$	–	0.01	0.98 ± 0.002	0.07 ± 0.005	0.04 ± 0.001	25.9
	U-ILR (ours)	Z, $[-1, 1]$	–	0.01	0.98 ± 0.002	0.07 ± 0.004	0.04 ± 0.001	32.2
	C-ILR (ours)	Z, $[-1, 1]$	$\lambda: 0.99$	0.01	0.98 ± 0.002	0.07 ± 0.005	0.04 ± 0.002	52.3
Magic	LSM	$[-1, 1]$	–	0.01	0.82 ± 0.002	0.43 ± 0.003	0.10 ± 0.005	–
	BSM	$[-1, 1], Z$	–	0.001	0.86 ± 0.005	0.46 ± 0.002	0.17 ± 0.007	–
	GPD	Z	$\alpha_E: 0.1$	0.01	0.87 ± 0.003	0.33 ± 0.004	0.03 ± 0.002	126
	SVGPC	$[-1, 1], Z$	–	0.01	0.86 ± 0.003	0.33 ± 0.004	0.03 ± 0.003	29.1
	U-ILR (ours)	Z	–	0.01	0.86 ± 0.003	0.33 ± 0.003	0.02 ± 0.004	25.3
	C-ILR (ours)	Z	$\lambda: 0.95$	0.01	0.87 ± 0.002	0.32 ± 0.005	0.02 ± 0.006	55.5
MiniBoo	LSM	Z	–	0.01	0.89 ± 0.003	0.27 ± 0.003	0.07 ± 0.004	–
	BSM	Z	–	0.001	0.90 ± 0.002	0.27 ± 0.003	0.08 ± 0.002	–
	GPD	Z	$\alpha_E: 0.1$	0.001	0.91 ± 0.003	0.23 ± 0.005	0.04 ± 0.003	5587
	SVGPC	Z	–	0.01	0.90 ± 0.001	0.23 ± 0.003	0.03 ± 0.001	45.8
	U-ILR (ours)	Z	–	0.01	0.90 ± 0.001	0.23 ± 0.003	0.03 ± 0.002	42.8
	C-ILR (ours)	Z	$\lambda: 0.95$	0.01, 0.001	0.91 ± 0.005	0.22 ± 0.010	0.03 ± 0.002	2535
Drive	LSM	$[-1, 1]$	–	0.01, 0.001	0.76 ± 0.014	0.69 ± 0.066	0.23 ± 0.022	–
	BSM	$[-1, 1]$	–	0.001	0.89 ± 0.026	0.66 ± 0.025	0.34 ± 0.015	–
	GPD	$[-1, 1]$	$\alpha_E: 0.0001$	0.01	0.99 ± 0.001	0.02 ± 0.002	0.05 ± 0.000	16839
	SVGPC	$[-1, 1]$	–	0.01	0.99 ± 0.002	0.03 ± 0.004	0.05 ± 0.001	492
	U-ILR (ours)	$[-1, 1]$	–	0.01	0.99 ± 0.001	0.03 ± 0.003	0.05 ± 0.001	495
	C-ILR (ours)	$[-1, 1]$	$\lambda: 0.999999$	0.01	0.99 ± 0.001	0.02 ± 0.002	0.05 ± 0.000	16099
letter	LSM	$[-1, 1]$	–	0.01	0.76 ± 0.004	1.47 ± 0.009	0.48 ± 0.004	–
	BSM	$[-1, 1]$	–	0.001	0.67 ± 0.012	1.24 ± 0.007	0.29 ± 0.012	–
	GPD	$[-1, 1]$	$\alpha_E: 0.0001, 0.001$	0.01	0.96 ± 0.002	0.12 ± 0.003	0.04 ± 0.004	4443
	SVGPC	$[-1, 1]$	–	0.01	0.96 ± 0.002	0.13 ± 0.008	0.04 ± 0.002	697
	U-ILR (ours)	$[-1, 1]$	–	0.01	0.96 ± 0.002	0.13 ± 0.008	0.04 ± 0.003	661
	C-ILR (ours)	$[-1, 1]$	$\lambda: 0.999999$	0.01	0.95 ± 0.002	0.16 ± 0.003	0.04 ± 0.001	4491
MoCap	LSM	Z, $[-1, 1]$	–	0.01	0.88 ± 0.062	0.58 ± 0.482	0.22 ± 0.146	–
	BSM	$[-1, 1]$	–	0.001	0.92 ± 0.003	0.53 ± 0.002	0.28 ± 0.002	–
	GPD	$[-1, 1]$	$\alpha_E: 0.001$	0.01	0.99 ± 0.001	0.05 ± 0.005	0.05 ± 0.001	5497
	SVGPC	$[-1, 1]$	–	0.01	0.97 ± 0.001	0.11 ± 0.005	0.04 ± 0.002	154
	U-ILR (ours)	$[-1, 1]$	–	0.01	0.97 ± 0.001	0.11 ± 0.003	0.05 ± 0.002	109
	C-ILR (ours)	$[-1, 1]$	$\lambda: 0.999$	0.01	0.99 ± 0.001	0.05 ± 0.005	0.05 ± 0.001	2738

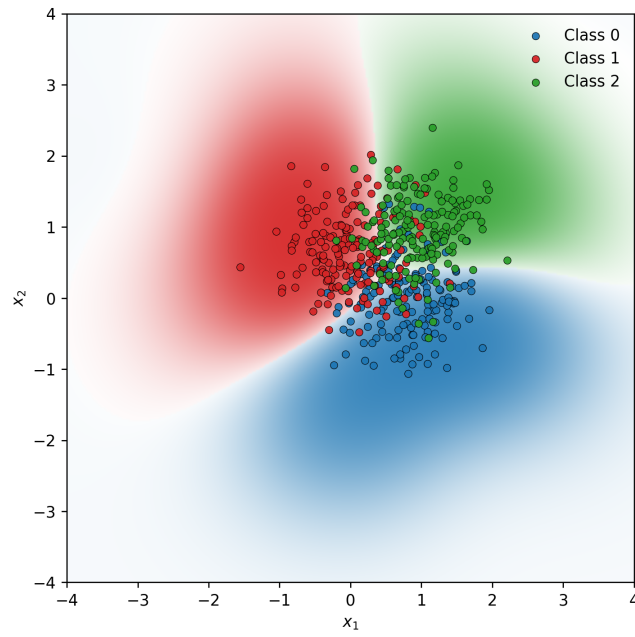


Figure D: Visualization of the Exact-ILR predictive probabilities on the same setup used in the experiment “Effect of the parameter λ ”, with $\lambda = 0.9$ and $s^2 = 0.5$. Training points are shown with their true class labels. The background color indicates the predicted class, and its intensity indicates the predicted probability. The model assigns high confidence near the training data and lower confidence away from the data, illustrating a reduction in class probability for out of distribution inputs.